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Studierichting: Informatica
Oefeningenreeks 5A nr. 6.4

$$\int_0^{\frac{\pi}{2}} \sin 2x dx$$

$$t = 2x \Rightarrow \frac{1}{2}dt = dx$$

$$x = 0 \Rightarrow t = 0$$

$$x = \frac{\pi}{2} \Rightarrow t = \pi$$

$$\int_0^{\frac{\pi}{2}} \sin 2x dx = \frac{1}{2}\int_0^{\pi} \sin t dt$$

$$= \frac{1}{2}\left[-\cos t\right]_0^{\pi}$$

$$= \frac{1}{2}\left(-\cos \pi + \cos 0\right)$$

$$= \frac{1}{2}\left(-(-1) + 1\right)$$

$$= \frac{1}{2} \cdot 2$$

$$= 1$$

$$\int_0^{\frac{\pi}{2}} \sin 2x dx = 1$$

References